

IEM 4723 Information Systems Design

Assignment 1: Git and Docker

Overview

This assignment combines the Git and GitHub workflow covered in Lectures 2 and 3 with the Docker skills covered in Lectures 3 and 4. You will fork a starter repository containing a template `Dockerfile` and a small example application, complete a series of Docker tasks on a dedicated branch, and submit your work as a Pull Request back to the course repository.

Git and GitHub Workflow [20 points]

1. Fork the starter repository at https://github.com/disys-lab/IEM4723_Fall2026_Assignments into your own GitHub account.
2. Clone your fork locally.
3. Create a new branch (any name of your choosing, for example `assignment1-work`) before making any changes. Do not commit directly to `main`.
4. Create a file named `docker_commands.sh` in your branch. As you complete each question below, append the exact commands you ran to this file under a comment header for that question (for example, `# Question 1`).
5. Complete Questions 1 through 4 below on this branch. Each question should correspond to at least one commit that updates `docker_commands.sh`, plus any other files that question asks you to add, with a clear, descriptive commit message.
6. Open a Pull Request from your branch back to the upstream course repository once you are finished.

Question 1: Build and Tag [20 points]

Using the provided `Dockerfile`:

1. In the directory `regression_container` you will find a Python file called `Regressor.py` and a `Dockerfile`. You do not need to modify the `Dockerfile` or the Python file.
2. Build a container image named `linear-regression-app` which contains the file `Regressor.py` using the sample `Dockerfile`.
3. Tag two versions of the image: `linear-regression-app:v0.1.1` and `linear-regression-app:v0.1.2`.
4. Docker Hub requires images to be namespaced under your username. Re-tag your image as `<dockerhub_username>/linear-regression-app:latest`, then push that to the remote Docker Hub repository.
5. Append your `docker build` and `docker tag` commands, and the `docker push` command, to `docker_commands.sh` under `# Question 1`, and commit.

Question 2: Volume Mounting [20 points]

1. In the directory `volume-mounting` you will find a folder called `data/` which should contain a file called `stream.csv`. Additionally there will be a Python file called `Regressor.py` (similar to Question 1) which will build the linear regression model and store the regression coefficients in the `data/` directory under the filename `regression_coefficients.csv`. There will also be a `Dockerfile` which will help you build the container needed for this assignment. You are not allowed to modify either the Python file or the `Dockerfile` for this assignment.
2. Construct a `Docker run` command so that the `data/` folder on the host is available inside the container and confirm that the file `regression_coefficients.csv` written by the application inside the container is visible on the host.
3. Append `docker run --volume` commands to `docker_commands.sh` under `# Question 2`. Commit this along with `regression_coefficients.csv`.

Question 3: Layering [20 points]

1. In the directory `layering` you will find a file called `LogisticRegressor.py` along with a folder called `data/` as well as a `Dockerfile` template.
2. Edit the existing `Dockerfile` with additional instructions such that it uses your `linear-regression-app:latest` image (from Question 1) as its base image.
3. Build this new image as `logistic_regression_layered:latest`, run it, such that that both the original application files from the base layer (Question 1) and your newly added file are present in the container.
4. Run the container to produce the logistic regression coefficients stored in `data/logistic_regression_coefficients.csv`. You would need to again mount the `data` directory through the volume mapping sequence followed in Question 2.
5. Append your `docker build` and `docker run` commands to `docker_commands.sh` under `# Question 3`, and commit. Your commit must include `data/logistic_regression_coefficients.csv`.

Question 4: Port Forwarding [20 points]

1. Consider the container `simple_file_server`, which lists the directories present inside the container. The code is provided in the `port_listening` folder. Build the container image.
2. The file server listens on port 8000 inside the container. Using port mapping, make it accessible from the host browser on port 3333. **The `Dockerfile` must remain unchanged.**
3. Confirm you can reach the application at `http://localhost:3333` and that the dataset `stream.csv` as the file available for download. Its fine if you see `data` and `Dockerfile` and `stream.csv` within the `data` directory.
4. Append your `docker run --publish` command(s) to `docker_commands.sh` under `# Question 4`, and commit.

Deliverables

Provide the following:

1. An open Pull Request from your branch into the upstream course repository.
2. A `docker_commands.sh` in your branch listing every command you ran, organized under a comment header per question.
3. The edited `Dockerfile` and `data/logistic_regression_coefficients.csv` from Question 3, and `regression_coefficients.csv` from Question 2.
4. A `README.md` in your branch containing:
 - a. Your Docker Hub repository link and image tags for Question 1.
 - b. A screenshot of your branch or network graph showing your branch history.

Grading

This assignment is graded out of 100 points as shown above, which accounts for 60% of your grade for this assignment. The remaining 40% comes from a short, three-question viva, where you will be asked to explain specific choices in your submission, such as your branching decisions or a particular command flag.